

B12 Series B

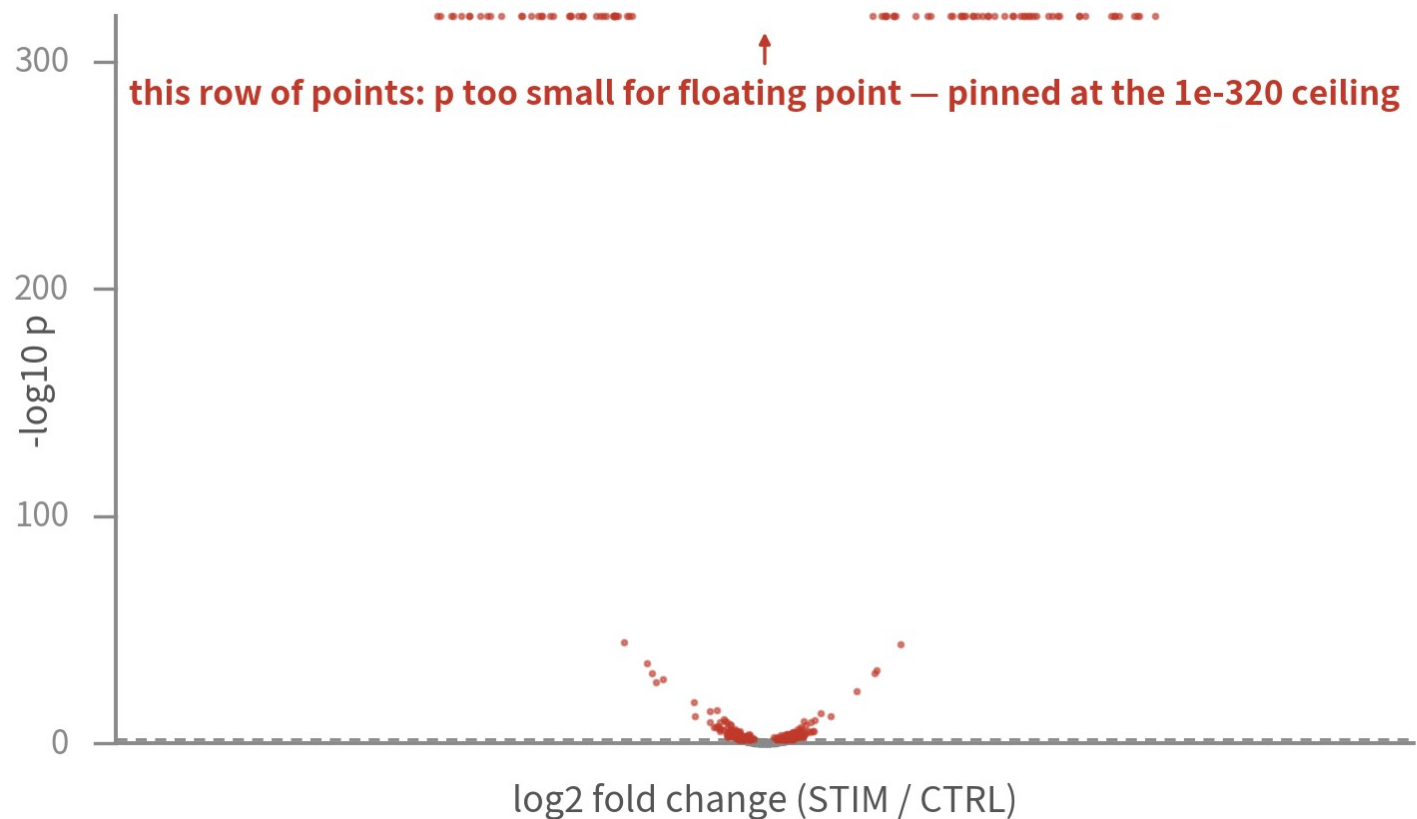
Differential expression done right: pseudobulk + DESeq2

Why FindMarkers across conditions hands you 3,000 'significant' genes — most of them statistical mirages.

About 40 min | Prerequisites: B11 | Data: ifnb (IFN-beta vs control) | Main-line B1-B12 finale

3,000 significant genes, p down to 1e-200

FindMarkers: 2,700 cells treated as replicates



simulated data: 8+8 samples, donor variation, 12% true DE

FindMarkers straight on stim vs ctrl: results too beautiful to be true — which is exactly the problem

significant genes

303 / 900

205 of them have no real effect

smallest p-value

~1e-320

beyond floating-point range

WHERE THIS EPISODE STARTS

**These p-values are not good —
they are impossibly good.**

In a 16-person experiment, $p = 1e-200$ means one thing: the test is counting the wrong unit.

What you will learn

- 1 Tell the two DE questions apart: cluster-vs-cluster markers vs same-cell-type across conditions
- 2 Explain how pseudoreplication inflates p-values — and watch it happen in simulation
- 3 Run the AggregateExpression → DESeq2 pipeline and read volcanoes and result tables correctly

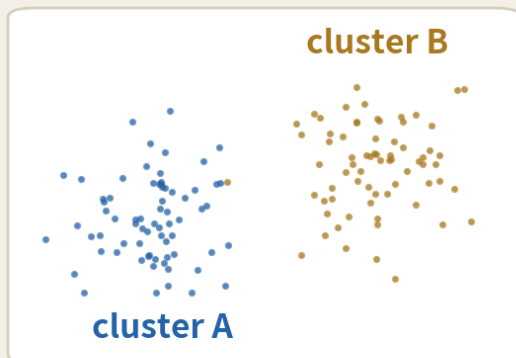
01

Which difference are you asking about?

Two similar-looking questions with different units of observation

Two DE questions, two toolkits

Q1: what is this cluster?

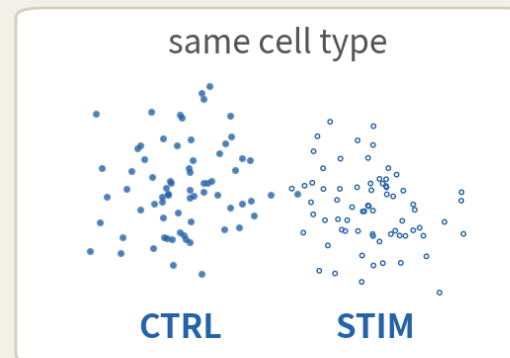


cluster A vs cluster B
(within the same samples)

FindMarkers is fine

done in B10: markers, naming

Q2: what did the stimulus change?



same cell type, CTRL vs STIM
(across samples and conditions)

pseudobulk + DESeq2

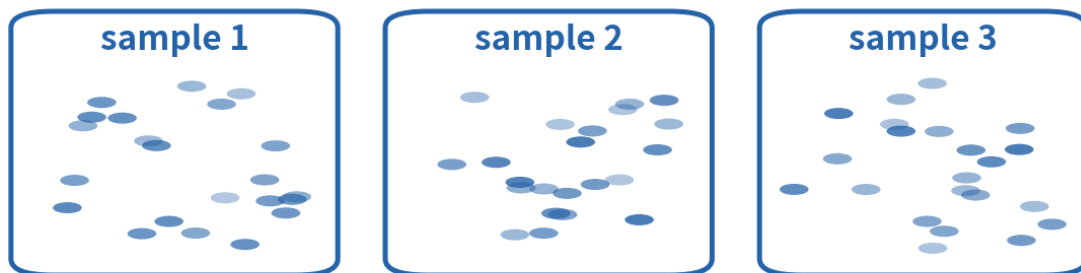
this episode: inference about treatment

the questions look alike, but the unit of observation differs — so do the tools

Marker finding was B10; this episode asks what the stimulus changed — inference about the treatment itself

The independent unit is the sample, not the cell

CTRL

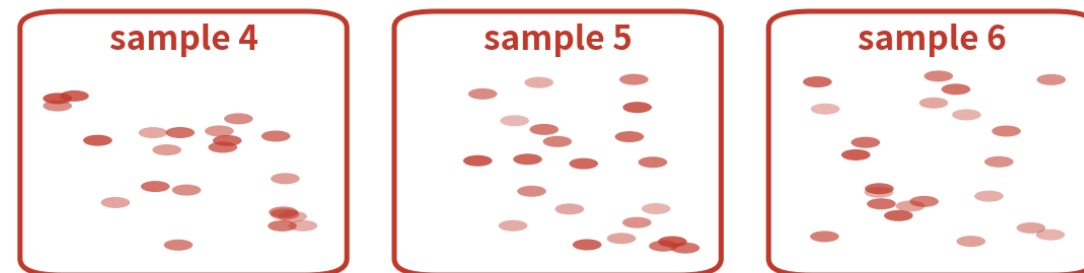


statistics counts biological replicates

$n = 3$ vs 3 (samples)

VS

STIM



cell-level tests count every cell

" $n = 12,000$ vs $13,000$ "?



cells from one sample share donor and batch effects — they are not independent

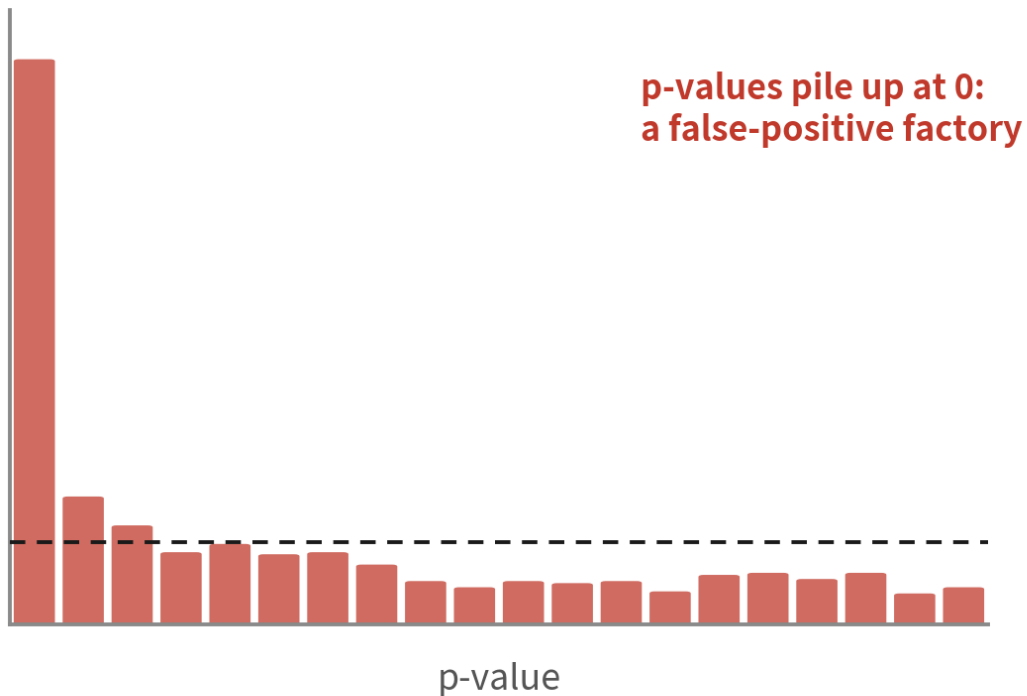
30,000 cells from one sample share donor and batch — counting them as n is pseudoreplication

Watch the p-values inflate

simulated data

cell-level wilcoxon (naive)

205 of 802 null genes called significant (after BH)

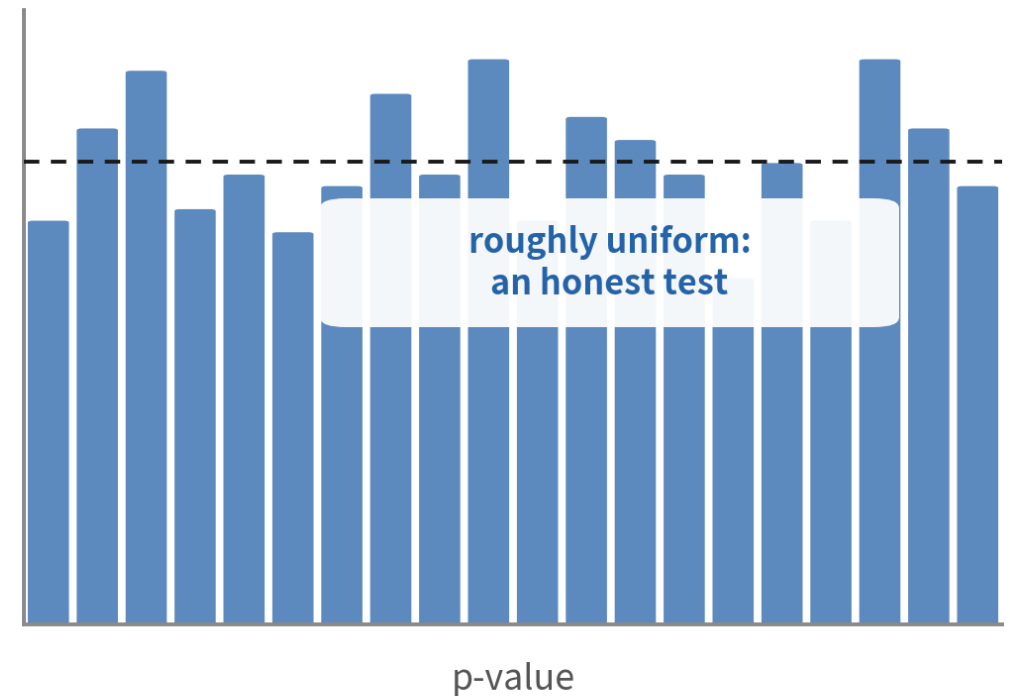


these genes have zero condition effect — ideal p-values hug the dashed line (uniform)

Simulation: genes with zero condition effect — the cell-level test still calls 200+ of them significant

pseudobulk (sample level)

9 of 802 null genes called significant (after BH)

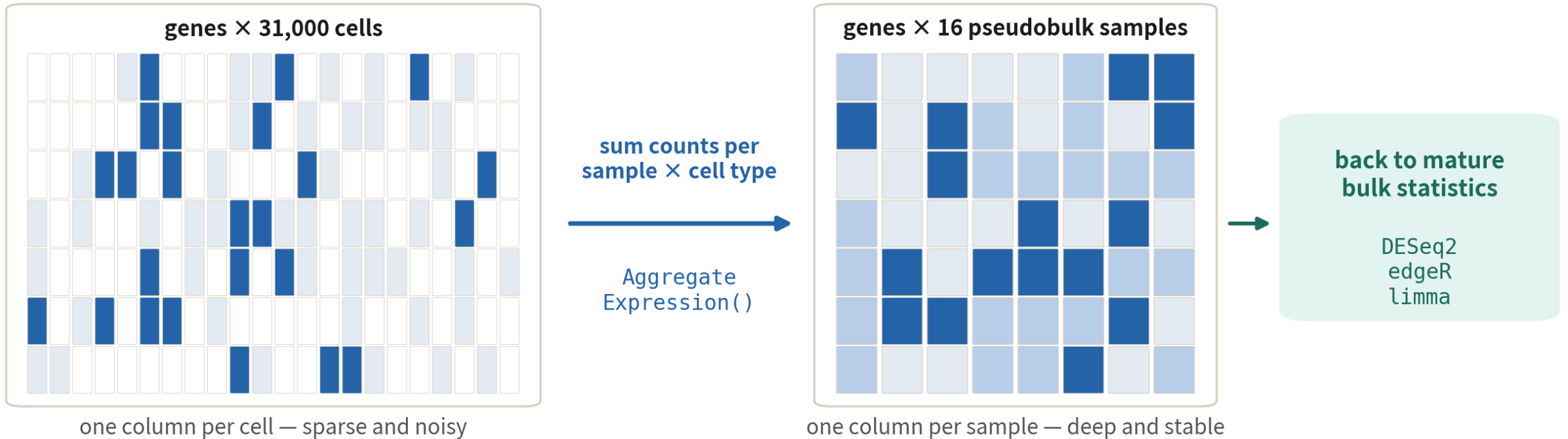


CORE IDEA

**Biological replicates are samples.
Cells are repeated measurements of the same
sample.**

A1 told the $n=1$ story. Thirty thousand cells do not turn $n=8$ into $n=30,000$.

Pseudobulk: back to one column per sample



summing absorbs dropout zeros and restores samples as the unit — two problems solved at once

Sum counts per sample × cell type, then hand the matrix to mature bulk statistics

Three problems pseudobulk solves at once

An honest n



the unit returns to
samples; p-values
return to believable
scales

Zeros absorbed



summed counts are no
longer sparse; dropout
needs no special
model

Statistics for free



a decade of DESeq2 /
edgeR methodology,
inherited directly

02

Load the data; run the naive way first

Keep this result — we will confront it later

From B11: load the integrated ifnb

```
library(Seurat)
set.seed(1234) # 全系列固定 seed

ifnb <- readRDS("output/ifnb_b11_integrated.rds")
ifnb <- JoinLayers(ifnb, assay = "RNA") # 把 split 的 layer 合回來
table(ifnb$stim)
```

CTRL STM
6548 7451

DE uses the RNA assay; JoinLayers merges the counts layers B11 had split

The naive way: FindMarkers across conditions

```
idents(ifnb) <- "seurat_annotations"  
m_ono <- subset(ifnb, idents = "CD14 Mono") # 先鎖定同一型細胞  
idents(m_ono) <- "stim "  
  
naive <- FindMarkers(m_ono, ident.1 = "STM ", ident.2 = "CTRL")  
head(naive, 3)  
sum(naive$p_val_adj < 0.05)
```

```
      p_val avg_log2FC pct.1 pct.2 p_val_adj  
ISG15      0      4.98 0.998 0.229      0  
IFI3       0      4.50 0.964 0.052      0  
IFI6       0      4.38 0.969 0.080      0  
[1] 3187
```

Default wilcoxon, one observation per cell — 3,187 significant genes, p flushed to zero

Write these three numbers down

3,187 significant



over a third of all
tested genes 'changed'

p_val = 0



too small for floating
point — not a real zero

**We will audit
this**



after pseudobulk, we
come back and watch
this list shrink

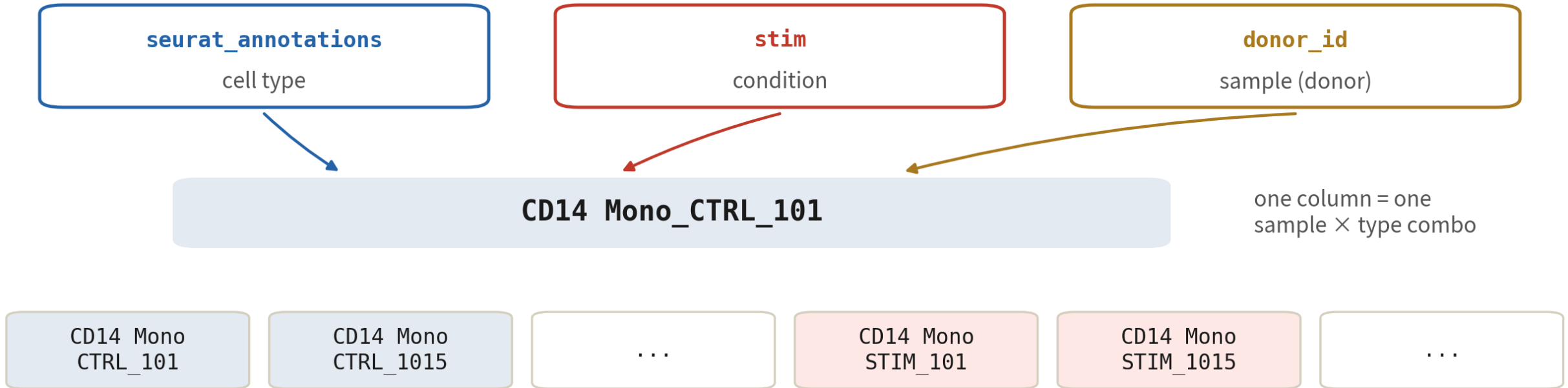
03

AggregateExpression: build the pseudobulk matrix

One column per sample $\times$ cell type combination

The three keys of group.by

```
group.by = c("seurat_annotations", "stim", "donor_id")
```



what gets summed is raw counts from the RNA assay — not data, scale.data, or integrated values

CD14 Mono: 8 CTRL + 8 STIM columns → 16 'samples' for DESeq2

Donor identities come from demuxlet (Kang et al. 2018); the R script includes download and matching code

Sum by cell type × condition × donor

```
# donor_id 欄位的建立見 R/B12_pseudobulk_de.R 第 2 節
pseudo <- AggregateExpression(ifnb, assays = "RNA",
                             group.by = c("seurat_annotations", "stim ", "donor_id"))
counts <- pseudo$RNA
dim (counts)
```

```
[1] 14053  198
```

14,053 genes × 198 columns: 13 cell types × 2 conditions × 8 donors (some combos empty)

Extract CD14 Mono and build the sample sheet

```
cd14 <- counts[,grep("^CD 14 Mono_", colnames(counts))]  
colnames(cd14)[1:2]  
parts <- strsplit(colnames(cd14), "_")  
coldata <- data.frame(  
  condition = factor(sapply(parts, `[`, 2),  
                     levels = c("CTRL", "STM")),  
  donor     = sapply(parts, `[`, 3),  
  row.names = colnames(cd14))  
table(coldata$condition)
```

```
[1] "CD 14 Mono_CTRL_101" "CD 14 Mono_CTRL_1015"  
CTRL STM  
8 8
```

16 pseudobulk samples: n = 8 vs 8 — the world DESeq2 is meant to see

THE RULE OF THIS SECTION

**Pseudobulk sums raw counts from the RNA
assay —
never any corrected values.**

Not integrated, not scale.data. B11's trap three plugs straight into this episode.

04

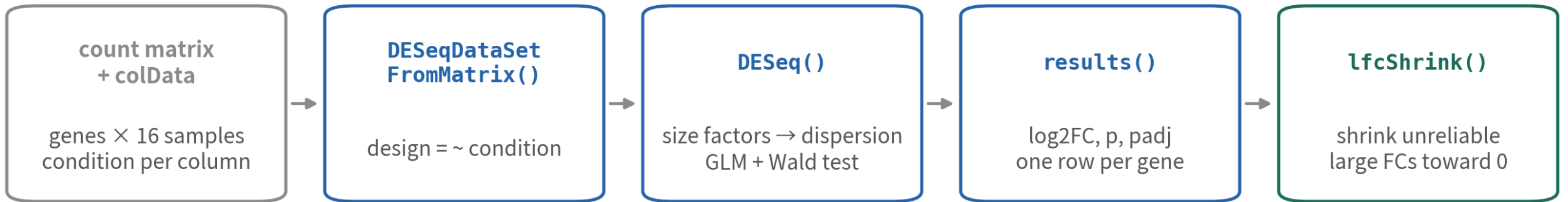
DESeq2: mature statistics in three calls

DESeqDataSetFromMatrix → DESeq → results

The DESeq2 pipeline

DESeq2: a decade of bulk RNA-seq statistics, inherited for free

negative binomial model + between-sample variance — exactly what cell-level tests lack



three lines of code end to end; the maths is worked through in Series C

A negative binomial model plus between-sample variance — the two pieces cell-level tests lack

Build the DESeqDataSet

```
library(DESeq2)
keep <- rowSums(cd14) >= 10      # 全部樣本加起來不到 10 個
                                # counts 的基因，先剔除
dds <- DESeqDataSetFromMatrix(countData = cd14[keep, ],
                              colData = colData,
                              design = ~ condition)

dds
```

```
class: DESeqDataSet
dim : 12446 16
colData names(2): condition donor
```

design = ~ condition: expression modelled by condition. The design is a model statement, not an option

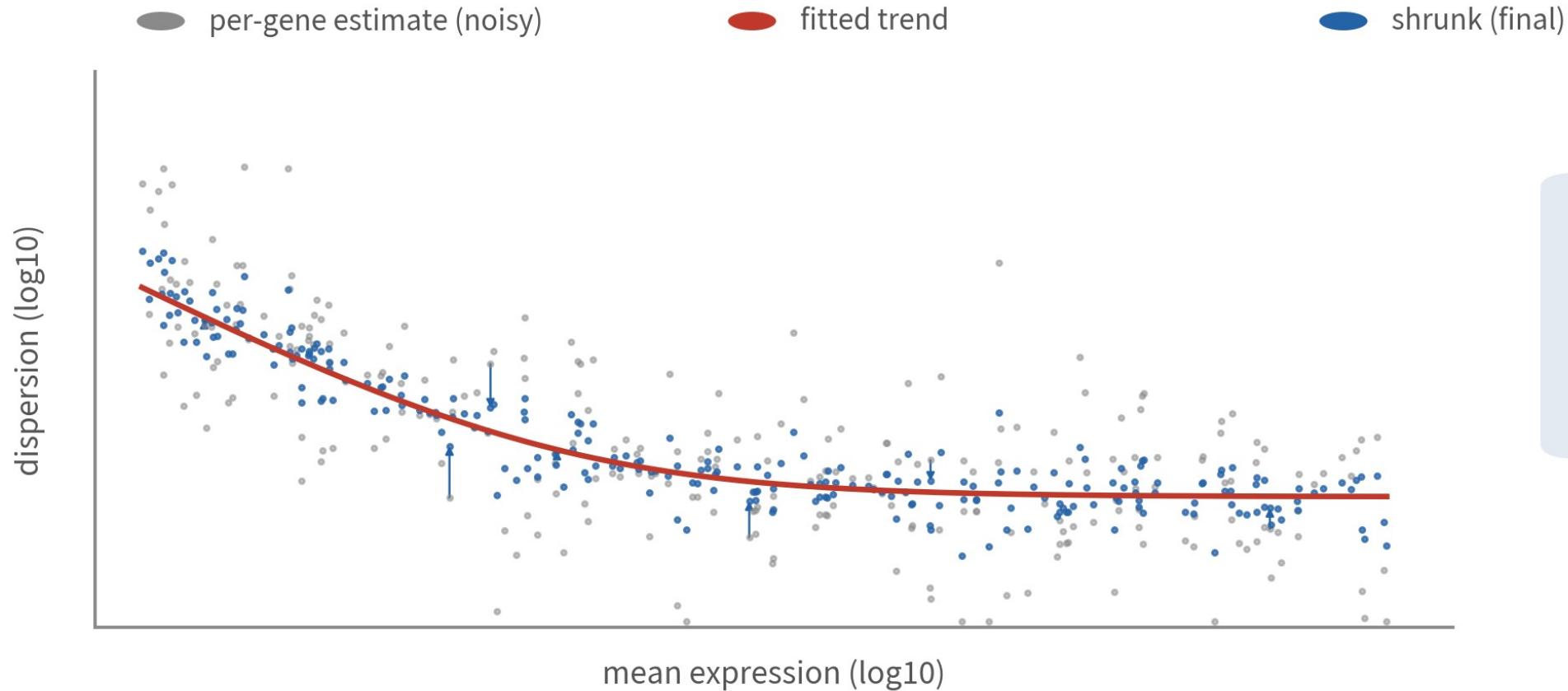
Estimation and testing in one call

```
dds <- DESeq(dds)
```

```
estimating size factors  
estimating dispersions  
gene-wise dispersion estimates  
mean-dispersion relationship  
final dispersion estimates  
fitting model and testing
```

These six log lines are the six pipeline steps; size factors and dispersions all happen here

Dispersion shrinkage: the small-n safety device



why shrink?

16 samples can't pin down per-gene variance; borrow strength from the global trend

simulated data

16 samples cannot pin down per-gene variance; borrowing strength from the trend stabilises the estimate

Pull the results, read the summary

```
res <- results(dds, contrast = c("condition", "STM ", "CTRL"),  
               alpha = 0.05)  
summary(res)
```

```
out of 12446 with nonzero total read count  
adjusted p-value < 0.05  
LFC > 0 (up)      : 1024, 8.2%  
LFC < 0 (down)    : 906, 7.3%  
low counts [2]    : 2413, 19%
```

About 1,930 up plus down — audit against the naive 3,187 and an order of magnitude of false positives falls away

The results table, column by column

one row of results() = one gene's full verdict

	baseMean	log2FoldChange	lfcSE	pvalue	padj
ISG15	6006.7	8.01	0.29	1.9e-165	2.1e-161
IFIT1	2929.5	8.75	0.38	1.1e-119	6.2e-116
IFIT3	3622.8	7.13	0.30	4.9e-125	3.7e-121
CXCL10	1274.2	6.42	0.41	3.5e-62	8.8e-59

↑
mean across samples:
is the gene expressed
enough to trust?

↑
effect size: log2 ratio
STIM over CTRL
(a star of the show)

↑
standard error
of the log2FC

↑
raw p-value
(never filter on this)

↑
BH-adjusted:
filter on this column

the reading triangle: padj gates, log2FC sizes the effect, baseMean checks abundance

The reading triangle: padj gates, log2FC sizes the effect, baseMean checks abundance

lfcShrink: rein in unreliable fold changes

```
res.shr <- lfcShrink(dds, coef = "condition_STM_vs_CTRL",  
                    type = "apeglm")  
head(res.shr[order(res.shr$padj), ], 3)
```

	baseMean	log2FoldChange	lfcSE	pvalue	padj
ISG15	6006.7	8.01	0.293	1.9e-165	2.1e-161
IFIT1	2929.5	8.75	0.375	1.1e-119	6.2e-116
IFIT3	3622.8	7.13	0.300	4.9e-125	3.7e-121

Wild FCs from low-count genes shrink toward 0; interferon-stimulated ISG15 / IFIT genes top the table

What this section did

One dds object



counts, sample sheet,
and design in one

One results table



12,446 genes, one
verdict per row

One habit



report shrunk
log2FCs — ranking no
longer hijacked by
noise

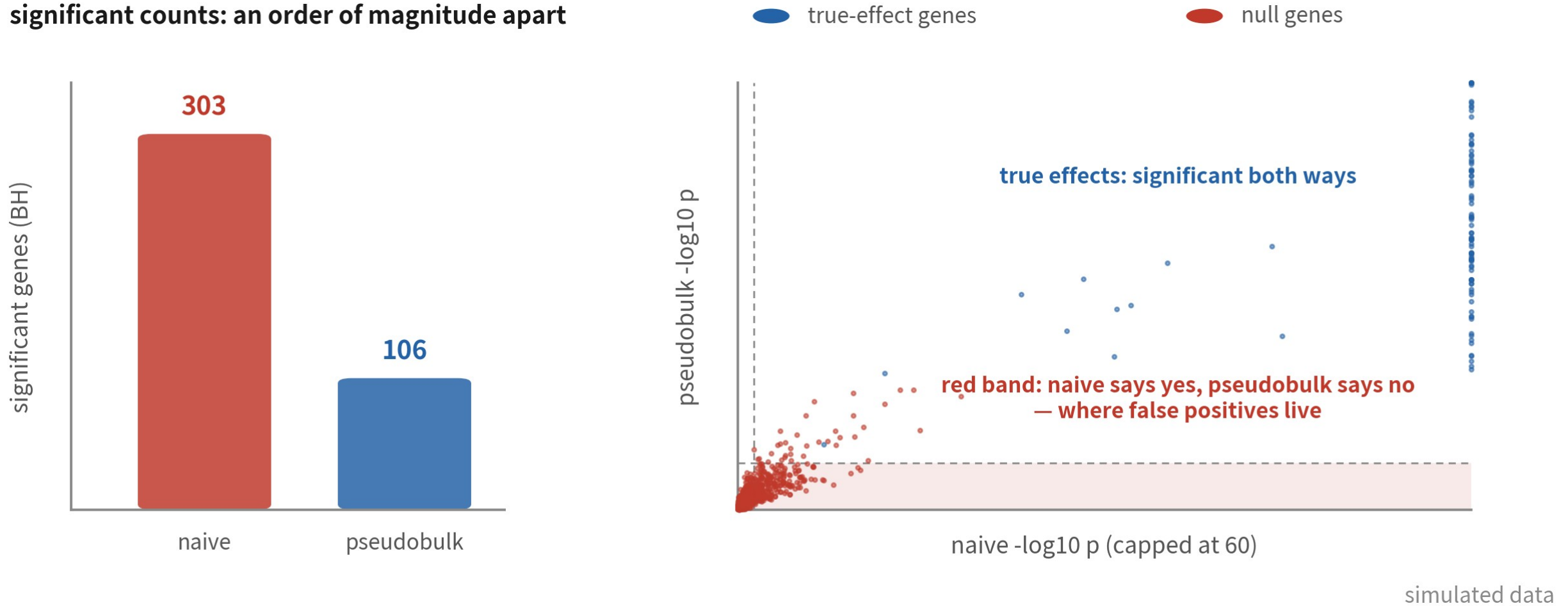
05

The audit, and how to read the answer

Naive vs pseudobulk — then turn results into figures

The audit: an order of magnitude apart

significant counts: an order of magnitude apart



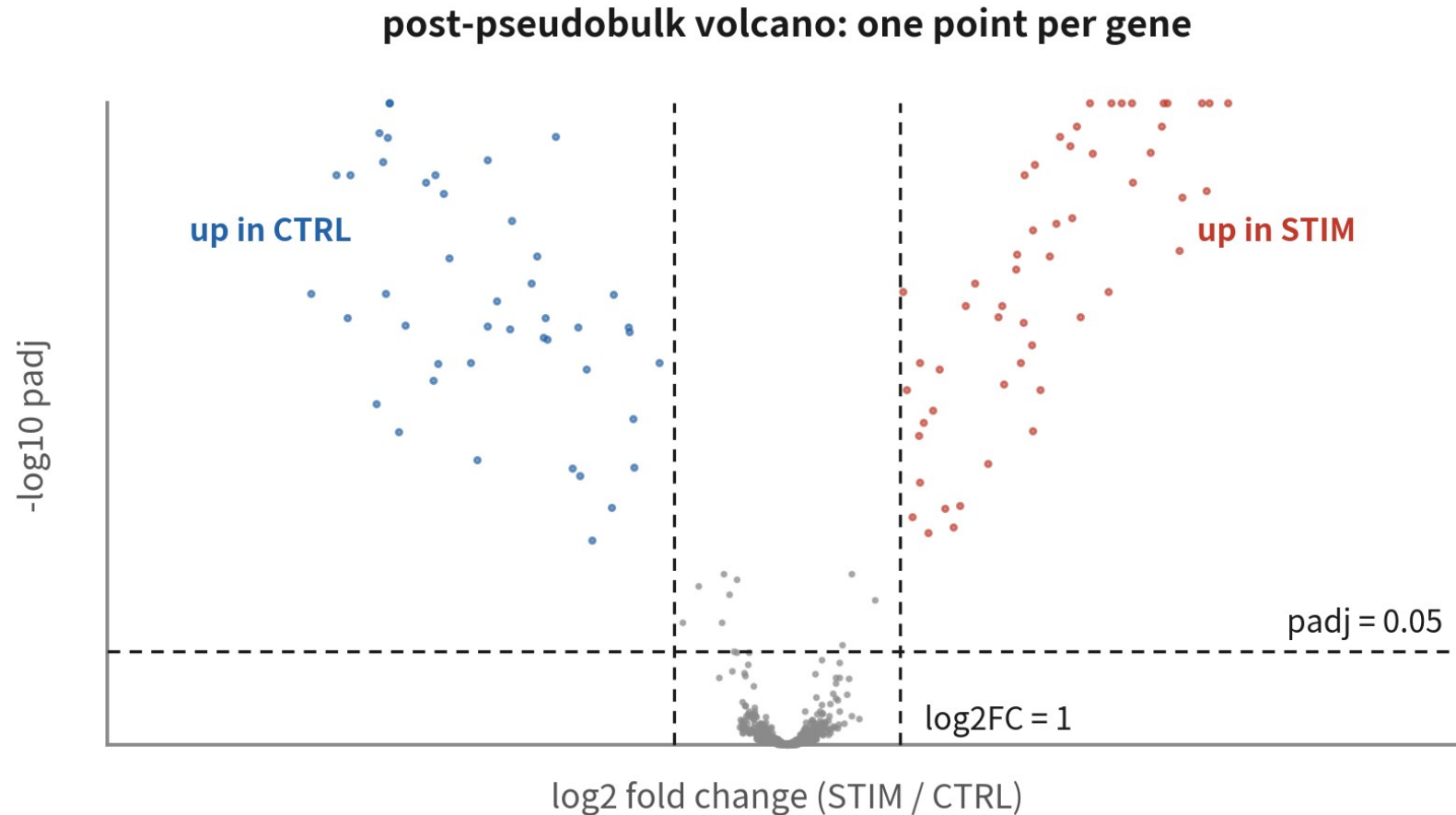
Genes on the red band — naive says yes, pseudobulk says no — are mostly donor variation, not stimulation

The volcano: ten lines of ggplot

```
library(ggplot2)
df <- as.data.frame(res.shr)
df$sig <- !is.na(df$padj) & df$padj < 0.05 &
  abs(df$log2FoldChange) > 1
ggplot(df, aes(log2FoldChange, -log10(padj), colour = sig)) +
  geom_point(size = 0.6) +
  scale_colour_manual(values = c("grey70", "#C0392B")) +
  geom_vline(xintercept = c(-1, 1), linetype = 2) +
  geom_hline(yintercept = -log10(0.05), linetype = 2) +
  theme_classic()
```

Significance is defined in code: $\text{padj} < 0.05$ AND $|\log_2\text{FC}| > 1$ — both gates must pass

How to read the volcano



report all three

- padj cutoff (0.05)
- $\log_2 \text{FC}$ cutoff ($|\text{FC}| > 1$)
- up / down gene counts

simulated data

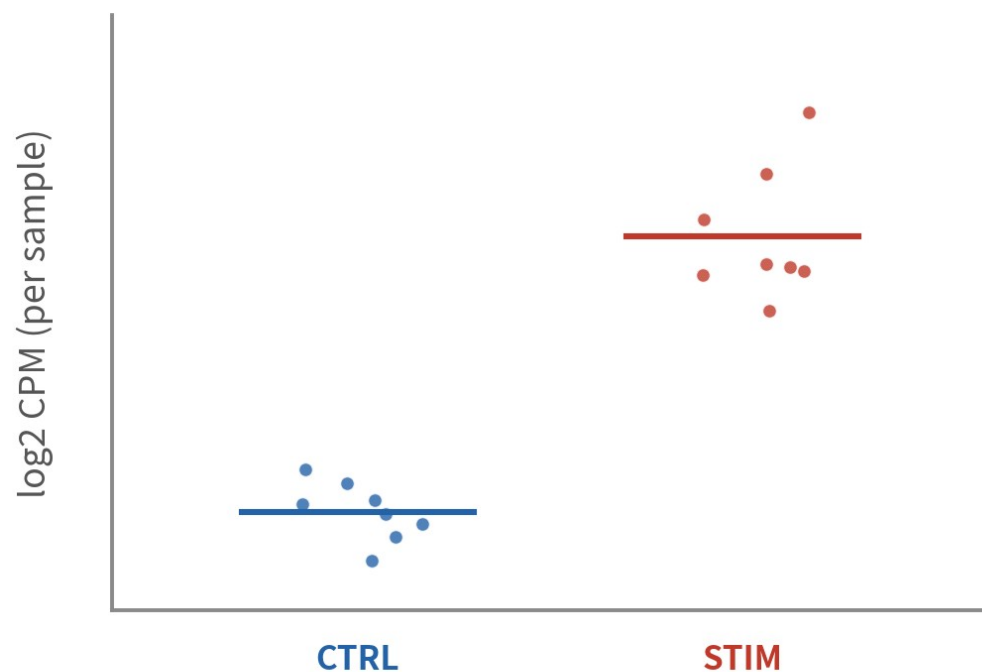
One point per gene; red top-right induced, blue top-left suppressed; report all three cutoffs

The final gate: one dot per sample

simulated data

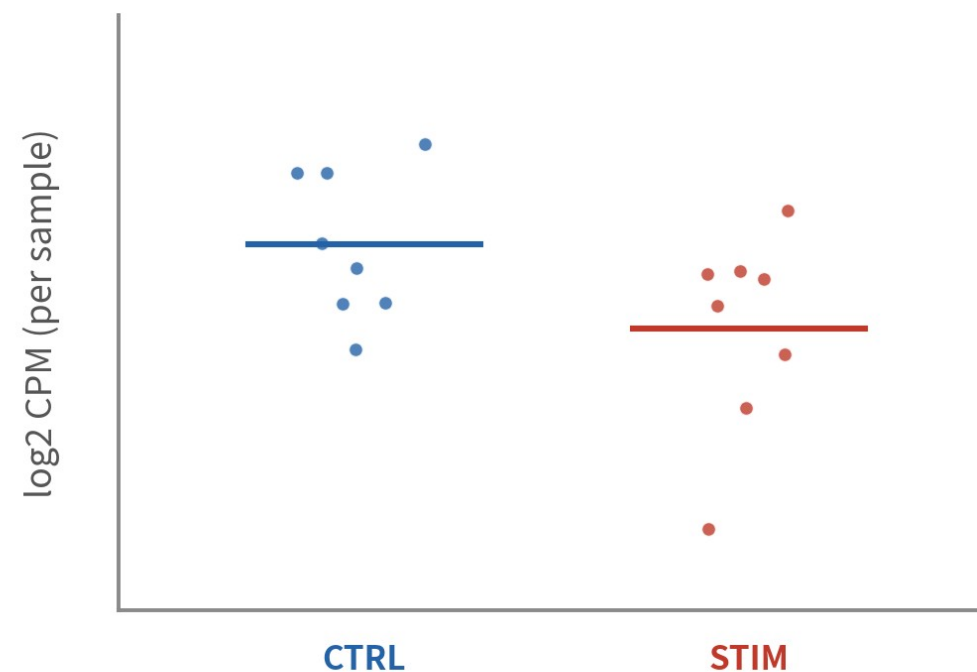
real: consistent across samples

naive $p \sim 1e-320$ pseudobulk $p = 9e-07$



fake: samples do not separate

naive $p \sim 1e-10$ pseudobulk $p = 0.079$



the final gate before publishing: plot one dot per sample — the effect must hold at sample level

Real effects hold at sample level; fake ones fall apart here — plot every headline gene before submitting

FOR YOUR METHODS SECTION

"Counts were summed per sample and cell type, tested with DESeq2; BH-adjusted $p_{adj} < 0.05$ and $|\log_2FC| > 1$."

Unit, tool, correction, cutoffs — the four things a reviewer asks for, in one sentence.

06

Where people get hurt

Three traps — each demonstrated for real

Trap 1: DE on integrated values

What it is

Fresh from B11, feeding the integrated assay straight into DE.



Why it happens

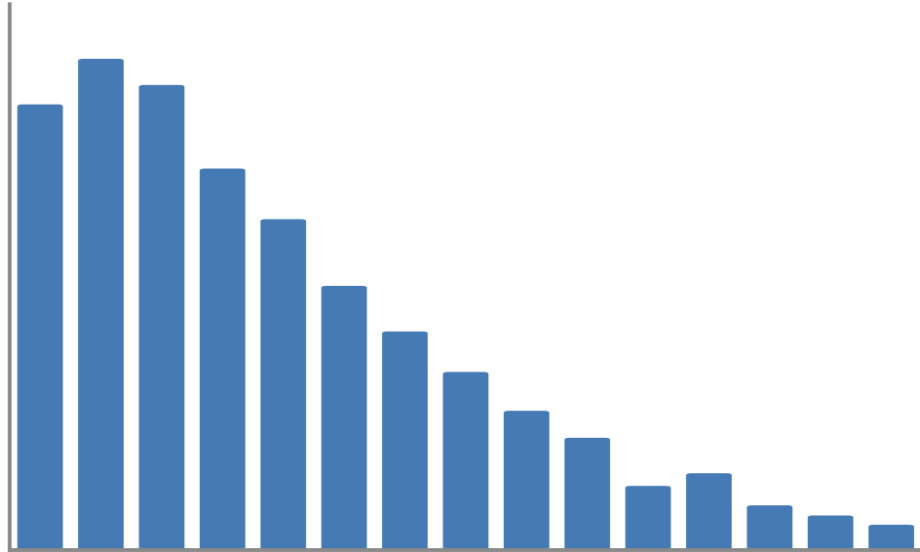
'The integrated data is cleaner' sounds perfectly reasonable.

What it costs you

Corrected values erase condition signal along with batch, go negative, aren't integers — DESeq2 errors out, and forcing it fabricates signal.

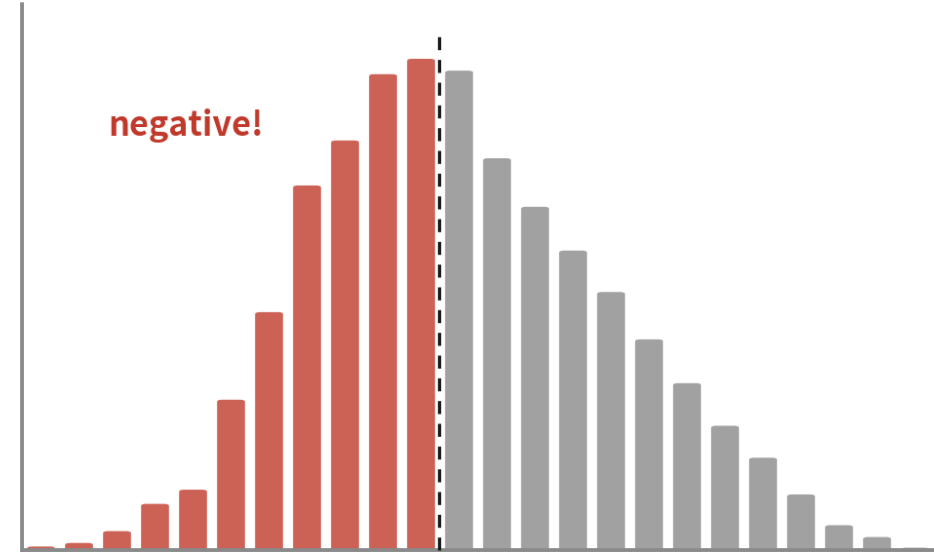
Counts and corrected values look nothing alike

counts in the RNA layer



counts: integers, non-negative — what DESeq2 models

the integrated "expression"



corrected values: continuous, can be negative; batch and some condition signal already erased

integrated values are coordinates for embedding, not expression — DE always goes back to RNA counts

Integrated values are embedding coordinates, not expression — DE always returns to RNA counts

Trap 2: forcing DE without biological replicates

What it is

Aggregating by condition only — one CTRL column, one STIM column, $n = 1$ vs 1 — and testing anyway.



Why it happens

'Pseudobulk collapses cells into samples anyway' — so why not collapse all the way?

What it costs you

Within-group variance is inestimable and DESeq2 refuses to run; ifnb only works because demuxlet recovers 8 donors — your own n must exist at design time.

n comes from design, not from computation

ifnb at face value: $n = 1$ vs 1



1 biological replicate per condition

one CTRL run, one STIM run
within-group variance inestimable
DESeq2 refuses to run

with demuxlet donors: 8 vs 8



8 biological replicates per condition

8 donors multiplexed per run
demuxed by genetic variants
what this episode does

your experiment: decide n at design time



3 biological replicates per condition

at least 3 biological replicates
per condition — you cannot add n
after sequencing

pseudobulk cannot conjure replicates: no statistic recovers an n your design never had

At least three biological replicates per condition — no statistic recovers an n the design never had

Trap 3: reading volcanoes by p alone

What it is

Sort the table by p_{adj} and ship the top ten straight into the paper.



Why it happens

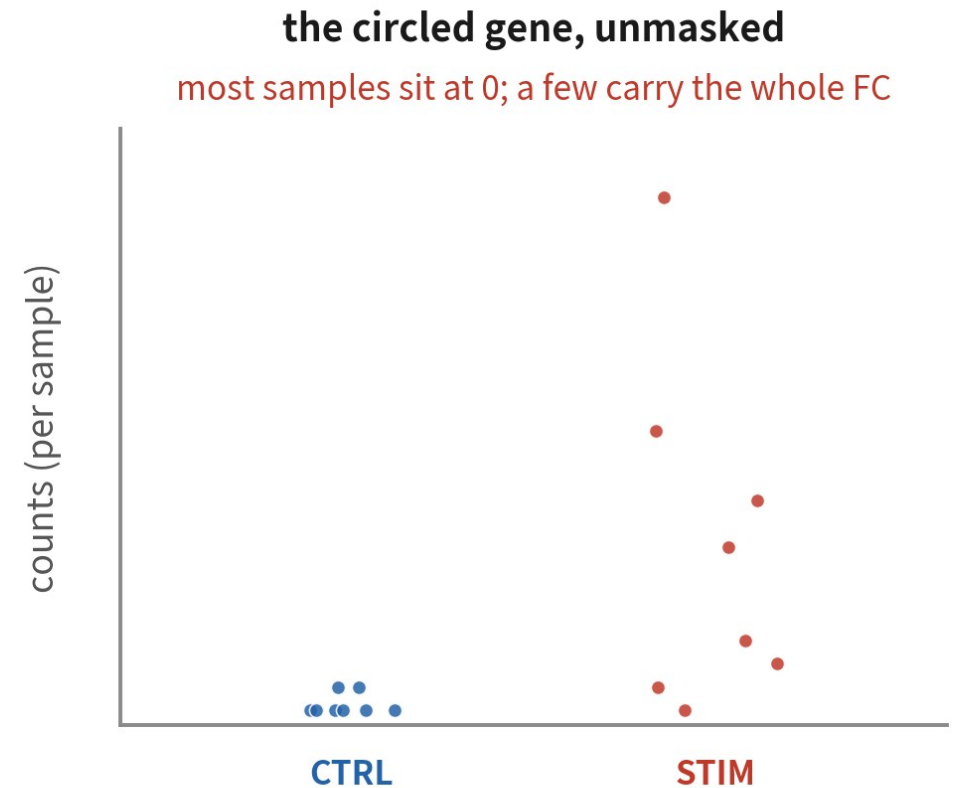
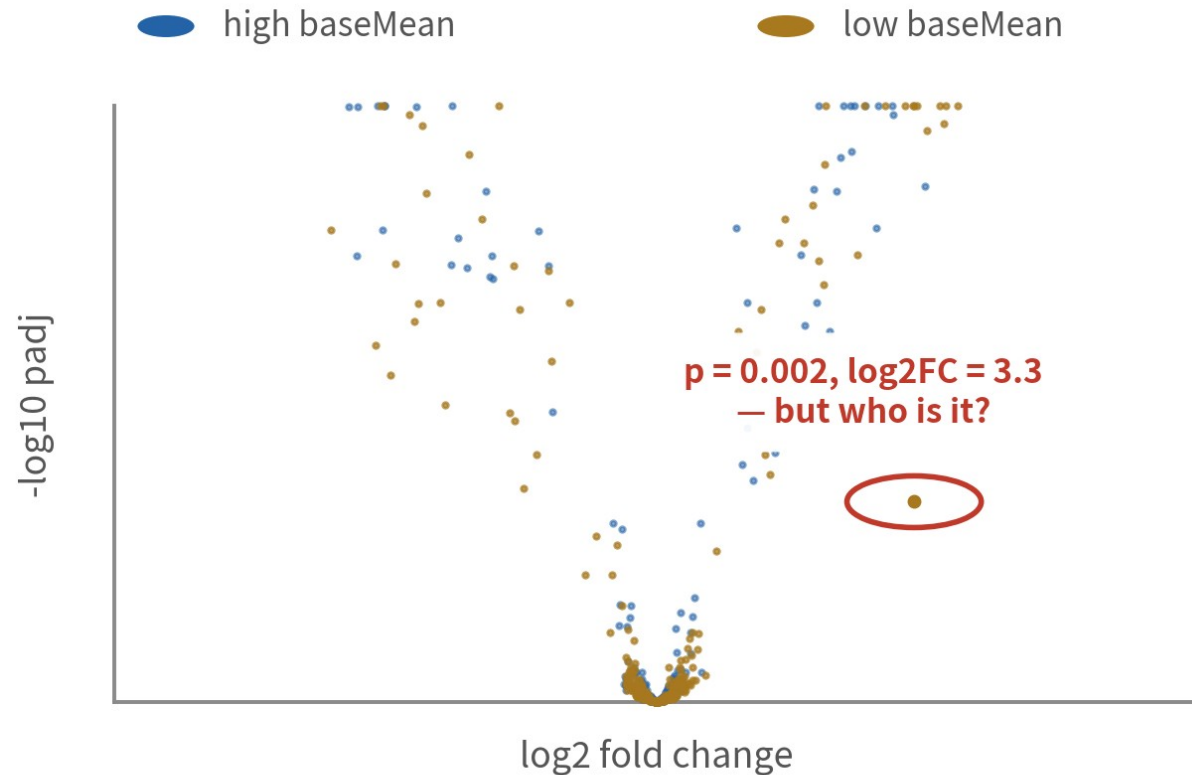
'Smallest p means most important' — an instinct carried over from Stats 101.

What it costs you

Low-count, high-variance genes sneak in: two or three samples carry the whole fold change, and it vanishes in the next cohort.

A $p = 0.002$ gene can look ugly up close

simulated data



low-expression genes double on a whisper of noise — check baseMean and per-sample counts before believing any FC

Check baseMean and per-sample counts before believing any FC — one noisy sample doubles a low-count gene

TRAPS, SUMMED UP

Three gatekeepers of DE: RNA counts, a real n , effect size.

Right data layer, honest sample count, effect size inspected — only then do p-values get a vote.

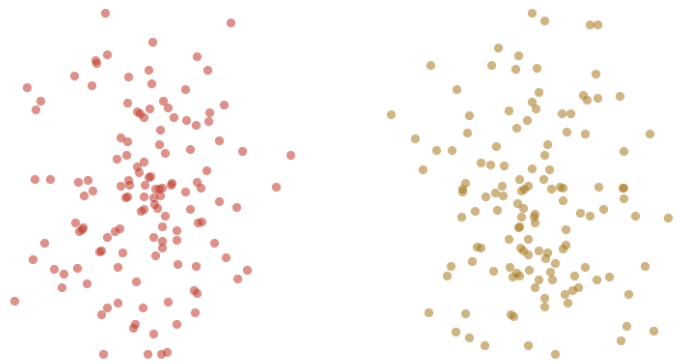
06+

Strategy Room: Complex Samples

The same pseudobulk, on a multi-patient tumour's core vs margin — which decisions change, and why

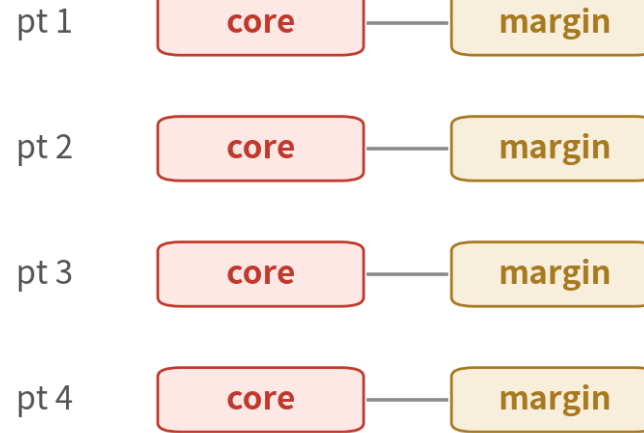
The patient is the unit — tumours magnify pseudoreplication

what the cell-level test thinks: $n = 2,400$



wilcoxon: 3,486 'significant'

the real design: $n = 4$ patient pairs



pseudobulk + paired DESeq2: 608

why tumours make it worse

patient 1's malignant cells



patient 2's malignant cells



cells within a patient share one CNV set —
more correlated than PBMC;
counting cells as n inflates worse

schematic; significant-gene counts from a real GSE84465 immune-cell core-vs-margin run

One patient's malignant cells share one CNV profile — more correlated than PBMC cells; n is 4 patient pairs, not 2,400 cells

Each type on its own, plus two gates

Field lessons from the four-patient GBM core-vs-margin data (GSE84465)

Mixed together = composition



DE on all cells tops out at MBP / PLP1 — margin just has more oligodendrocytes; split by type, MBP goes flat

MIN_CELLS = 20



each patient × site needs at least 20 cells to count as one pseudobulk sample — a 'bulk' summed from three cells is noise

MIN_PAIRS = 2



at least two patients must pass at both sites; in that GBM data only immune cells qualified — the margins held just 31 malignant cells in total

The statistical dividend of pairing

Core vs margin within one patient: the design grows from $\sim$ condition into $\sim$ patient + tissue

Across patients

Patient noise fills the error term

- GBM patients differ hugely — B11 showed malignant cells forming per-patient islands
- site effects wrestle patient variance; medium effects are nearly invisible
- needs far more patients than a paired design

Paired within patient

Absorb the patient, compare sites

- design = $\sim$ patient + tissue — the patient term soaks up individual differences
- 4 pairs already catch large effects — 4 immune-cell pairs yielded 608 significant genes
- medium effects still want ten-plus pairs — one more patient beats ten thousand more cells

When pseudobulk is impossible: three lines of defence

Below MIN_PAIRS you fall back to cell level — positioned as hypothesis-generating, never as conclusions

Defence	How	What it catches
1. Patient as covariate	MAST with latent.vars = patient; compare against default wilcoxon	genes only on the wilcoxon list = patient differences in disguise (e.g. XIST — really just sex)
2. Per-patient consistency	compute logFC per patient for top genes; keep only direction-consistent ones	'differences' carried by one patient; a column of NA visualises why pseudobulk failed
3. Label permutation	shuffle condition labels within patients and re-test (≥ 100 rounds); compare the real count	real signal should dwarf permuted counts; if patient and condition nearly coincide, the permutation itself has no power

Three log2FCs — when to report which

One DESeq2 result carries three effect sizes — each has its post; do not mix them

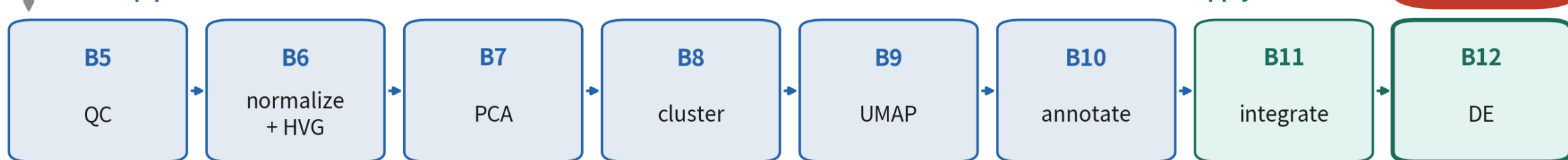
Which one	Nature	Where it goes
Unshrunk log2FC	straight from results(); low-count genes carry big FCs with big SEs	the volcano's x-axis — the volcano wants the raw shape
The Wald stat	$\log_2\text{FC} \div \text{SE}$ — effect and uncertainty in one number, almost no ties	the ranking statistic for GSEA (B13 uses it)
Shrunk log2FC	stable at small n: keeps real large effects, damps the noise	reported effect sizes and validation picks — the number for your tables
Choosing the shrinker	with only 8 samples (4 pairs) apegm's prior overwhelms the data — LFCs collapse into one column	16 samples here suit apegm fine; tumour pairings of 3–5 switch to ashr

The Series B map: the main line is complete

upstream B1–B4: FASTQ to count matrix



core pipeline B5–B10



apply B11–B12

you are here

up next

DE hands you a gene list — the advanced track answers 'now what':



main line B1–B12 complete; pick B13–B19 as needed (B19 is the cross-cutting strategy capstone; suggested B17→B19→B18) — the series closes at B18

The nineteen-episode map: main line B1–B12 complete; advanced B13–B19 are yours to pick (B19 is the cross-cutting strategy capstone)

ONE-LINE SUMMARY

**Sum cells back into samples —
and hand the statistics back to samples.**

Pseudobulk is not a new method; it is a translator that restores the correct statistical unit.

Check yourself 1

Comparing STIM vs CTRL within CD14 Mono cells — which approach is right?

- A Default FindMarkers wilcoxon, cells as observations
- B AggregateExpression then DESeq2
- C FindMarkers on the integrated assay
- D Subsample 100 cells per group, then wilcoxon

Answer B. Across conditions the unit is the sample: sum RNA counts per sample $\times$ cell type, then test with DESeq2. A and D both count cells as n — subsampling shrinks the inflated n but keeps the wrong unit; C uses corrected values, the wrong data layer.

Check yourself 2

Why do cell-level tests inflate p-values to $1e-200$?

- A The wilcoxon test is buggy
- B Sequencing depth is too shallow
- C Correlated cells from the same sample are treated as thousands of independent observations
- D Multiple-testing correction is too lenient

Answer C. Power explodes with n , but cells share donor and batch effects — the effective n is 8 vs 8. Donor-level variation gets misread as a condition effect, handing astronomical p-values even to null genes. A, B and D never touch the real issue: the wrong unit.

Check yourself 3

A pseudobulk hit: $\text{padj} = 1\text{e-}8$, $\log_2\text{FC} = 3.9$, $\text{baseMean} = 2.1$. Next move?

- A Big and significant — straight into the results
- B Check per-sample counts: a low-count gene's big FC may rest on a few samples
- C padj is small enough; ignore the other columns
- D Raise the $\log_2\text{FC}$ cutoff to 4 and refilter

Answer B. baseMean 2.1 means about two counts per sample — with a tiny denominator, one noisy sample yields $\log_2\text{FC}$ 3.9. Plot all 16 samples: if a couple of them carry the effect, this gene is no headline. A and C are exactly trap three; D papers over the problem with another arbitrary cutoff.

MAIN LINE COMPLETE • NEXT: B13

The main line ends; the road goes on

DE hands you a gene list — next, B13 answers: so what happened, biologically? Functional enrichment and gene set analysis.

Then B14 communication, B15 trajectories, B16 inferCNV and B17 deconvolution as needed; the series closes at B18. See you in B13.